EASTERN MEDITERRANEAN UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING

Course Code:	MENG222
Course Name:	Mechanics of Material
Semester/ Year:	Spring 2023-2024
Instructor:	Prof. Cafar Kizilors
Assistant:	Garshasp Keyvan Sarkon
Assessment:	Lab 1
Due Date:	One Week After Demonstration

LAB NAME: Torsion test

STUDENT(S) NO.: 23704019 **NAME(S), SURNAME(S): TARIK TARIQ**

SECTIONS:

No.	STUDENT OUTCOMES	COURSE LEARNING OUTCOMES	WEIGHT OF SECTION /100	MARKS OBTAINED	REMARKS
Report Presentation	1, 6	3, 4, 5	20		
Introduction and Procedure	1, 6	3, 4, 5	20		
Results and Calculations	1, 6	3, 4, 5	30		
Discussion & Conclusion	1, 6	3, 4, 5	30		
TOTAL: (Out of 100)					

	Course Learning Outcomes	Student Outcomes 1	2	3	4	5	6	7
1	To understand the basic principles of mechanics.	X						
2	To develop an ability to draw free-body diagrams.	X						
3	To develop an ability to formulate and use the equilibrium equations.	X					X	
4	Use scalar and/or vector analytical techniques for analyzing forces in statically determinate structures.	X					X	
5	To develop an ability to identify the principles and equations that apply, and use them in solving static and dynamic problems.	X					X	
6	Understanding Planar kinematics and kinetics of rigid bodies	X						
7	Understand Newton's law in motion, and recognize different kinds of particle motions	X						
8	To develop an ability to develop equations of equilibrium and equations of motion, and solve these for forces and/or different motion parameters, such as displacement, velocity, and acceleration of particles, rigid bodies, and simple mechanical systems.	X						
	Weight of Student Outcomes	H						

Instructions:

A. **Report presentation:** Draw neat, labeled diagrams where necessary. Write relevant equations and your assumptions where necessary. Be clear and specific and include units. Give explanatory notes where necessary. Carry out each step explicitly. Provide references to the data/model/equations used. References should be from the textbook or from an authentic source.

B. **Submission:** Upload your report as a pdf file to the "Lab Work" section on TEAMS within the designated time. The file uploaded should be named as **[MENG231-Lab-Group No]**. No late submissions will be accommodated.

ABSTRACT

The torsion test is an essential method for assessing the response of materials to twisting forces, assisting in the selection of materials for components exposed to such stresses in machinery. This report presents an analysis of a torsion test conducted using the NJS-20 Torsion Testing Machine. The experimental procedure involved preparing the specimen, setting up the apparatus, selecting test parameters, mounting the specimen, initiating the test, monitoring data, observing specimen behavior, and analyzing results. The collected data was utilized to calculate significant parameters like shear stress, shear strain, and modulus of rigidity. A comparison between the theoretical and experimental values of the modulus of rigidity revealed a substantial 96% discrepancy, highlighting the challenges in characterizing materials. Factors contributing to this variation include limitations in the experimental setup, assumptions made in theoretical models, and variations in the samples. To enhance accuracy, it is crucial to refine experimental techniques, validate theoretical assumptions, increase sample sizes, and conduct comprehensive error analyses. Despite these challenges, the torsion test remains invaluable for engineering purposes, providing guidance in material selection and design decisions.

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1. INTRODUCTION

The torsion test is a mechanical evaluation method wherein materials are subjected to torque to observe their response. This test holds significant importance for engineering purposes. It aids in the selection of appropriate materials for manufacturing components, particularly those exposed to torsional forces within machines. By twisting a standardized specimen and measuring the resulting angle of twist, we can determine various mechanical properties such as shear stress, shear strain, and ultimately, the modulus of rigidity. This modulus is a fundamental measure of a material's resistance to deformation under shear stress.

2. Description of apparatus

The NJS-20 Torsion Testing Machine is a comprehensive apparatus used to analyze how materials respond to torsional (twisting) forces. It consists of several essential components:

1. Base: The sturdy foundation of the machine that supports all other parts and ensures stability during testing.

2. Twisting Mechanism: This includes a wheel or knob that allows manual application of torque to the specimen. By turning the wheel, torsional force is applied to the material being tested.

3. Supporting Beams: These beams provide structural support to hold the specimen in place securely. They ensure that the material being tested remains stable and properly aligned during the torsion process.

4. Recorder: An integrated recording device that senses and measures the values of torque (the amount of twisting force applied) and angle of twist (the amount of rotation) at each moment during the test. This data is then displayed in real-time on a screen, allowing for accurate monitoring and analysis of the material's behavior.

Figure 1 NJS-20 TESTING MACHINE

3. THEORY

The derived relation between shear stress and torque applied, shear strain and angle of twist are:.

$\tau = Tr/J$

$\gamma = \Phi r/L$

Where J $= \pi/2 * r^4$, $\Phi = TL/JG$, *From hooks law*, $\tau = G\gamma$

τ = Shear stress (N/mm).

γ = Shear strain (radians).

T = Applied torque (N.m).

r = Radius of the specimen (mm).

J = Polar second moment of inertia (mm^4).

Φ = angle of twist (radians).

G = Modulus of rigidity (N/mm).

L = Gauge length (mm).

4. EXPERIMENTAL PROCEDURE

The experimental procedure for conducting a torsion test involves several steps to ensure accurate and reliable results:

1. Preparation of Specimen:

Measure the dimensions of the specimen accurately, including its length and diameter.

Ensure that the specimen's surface is clean and free from any defects or irregularities that could affect the test results.

2. Setup of Testing Apparatus:

Set up the torsion testing machine according to manufacturer guidelines.

Ensure that the machine is properly calibrated and functioning correctly.

3. Selection of Test Parameters:

Determine the direction of rotation and specify whether clockwise or counterclockwise rotation is required for the test.

Set the initial torque and angle of twist to zero on the testing machine.

4. Mounting the Specimen:

Securely clamp or mount the specimen onto the torsion testing machine, ensuring proper alignment with the twisting mechanism.

5. Initiation of Test:

Start the test by rotating the wheel or knob of the torsion testing machine at a constant speed.

Begin recording the values of applied torque and angle of twist displayed on the recorder's screen.

6. Monitoring and Data Collection:

Continuously monitor the test as torque and angle of twist values are displayed in real-time.

Record the data at regular intervals throughout the test duration.

7. Observation of Specimen Behavior:

Observe any changes in the specimen's appearance or behavior, such as deformation or failure, during the test.

8. Termination of Test:

Stop the test once the specimen fails or breaks, indicated by a rapid decrease in torque to zero.

Ensure that the testing machine is turned off safely and securely.

9. Data Analysis:

Analyze the recorded data to calculate important parameters such as shear stress, shear strain, and modulus of rigidity using appropriate equations and formulas.

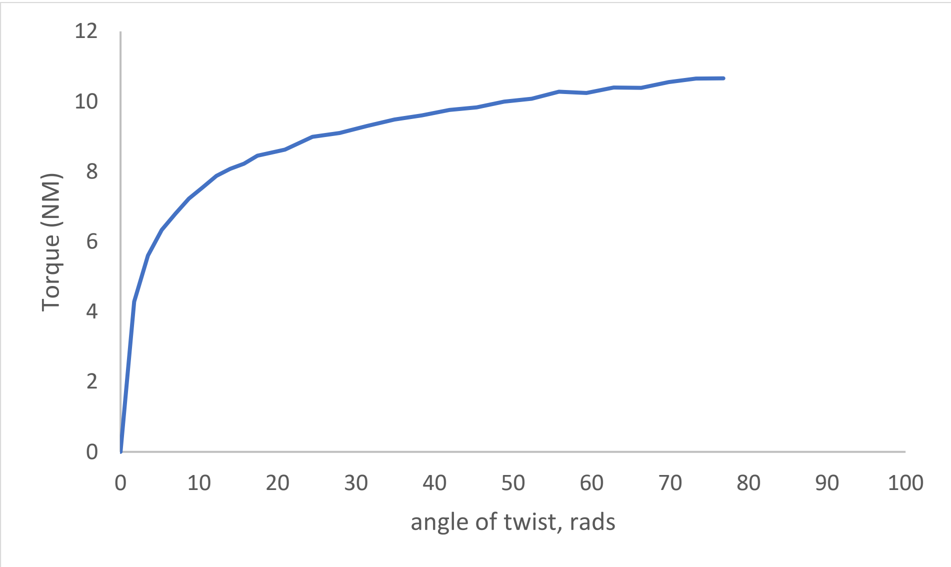
5. RESULTS AND DESCUSSION

The table below contains values of angle of twist in degrees and applied torque which were directly gotten from the experiment. The data was then uploaded on excel where angle of twist in degrees were converted to rads by using a factor of 2π/360. Lastly, the table also contains shear stress values

Table of results

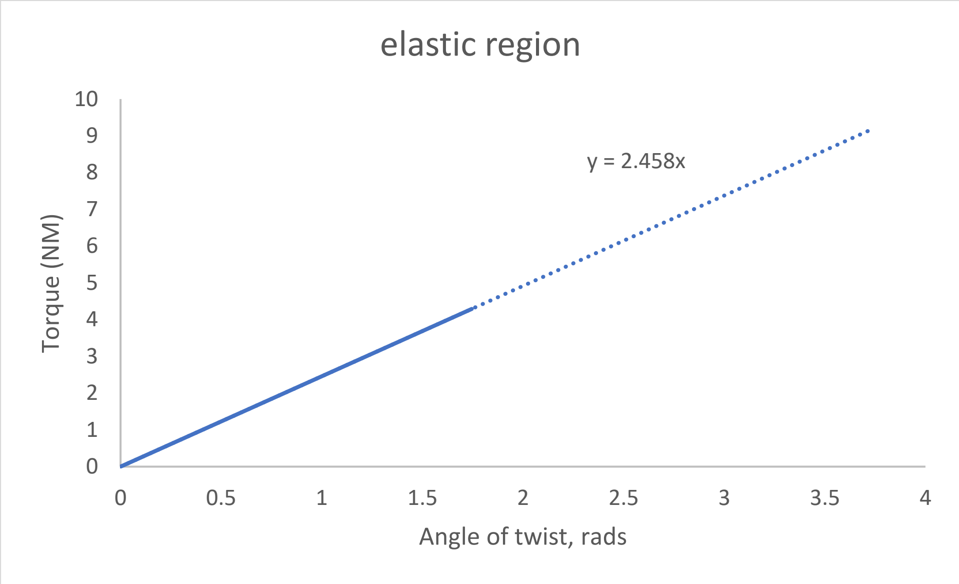
Angle of Twist	Angle of Twist	Applied Torque	Shear stress
(degree)	(rads)	NM	Mpa
0	0	0	0
100	1.7453293	4.29	174.79032
200	3.4906585	5.6	228.16453
300	5.2359878	6.33	257.9074
400	6.981317	6.8	277.05692
500	8.7266463	7.23	294.5767
600	10.471976	7.55	307.61467
700	12.217305	7.88	321.06008
800	13.962634	8.08	329.20882
900	15.707963	8.22	334.91293
1000	17.453293	8.45	344.28397
1200	20.943951	8.63	351.61783
1400	24.43461	8.99	366.28555
1600	27.925268	9.1	370.76736
1800	31.415927	9.3	378.91609
2000	34.906585	9.48	386.24995
2200	38.397244	9.6	391.13919
2400	41.887902	9.76	397.65817
2600	45.378561	9.83	400.51023

2800	48.869219	10	407.43665
3000	52.359878	10.08	410.69615
3200	55.850536	10.28	418.84488
3400	59.341195	10.24	417.21513
3600	62.831853	10.4	423.73412
3800	66.322512	10.39	423.32668
4000	69.81317	10.55	429.84567
4200	73.303829	10.65	433.92004
4400	76.794487	10.66	434.32747



Graph 1 torque vs angle of twist

By using the graph 1 and 2, some values were determined:
Torque at the limit of proportionality = **4.29 N.m**.
Angle of twist at the limit of proportionality = **1.7453293 radians**.
Shear stress at the limit of proportionality = **174.79032 MPa.**
Ultimate shear stress = **434.32747 MPa.**
Fracture shear stress = **434.32747 MPa.**



Graph 2 torque vs angle of twist (elastic region)

$slope = 2.458\ N.m/rad$

$G(Modulus\ of\ rigidity) =$ (T/θ) * (L/J) = slope x (L/J) = $2.458\ N.m/rad$ x (0.0762m/6.13592 x 10^-11) = 305,249,516.1pa/10^9 = 3.0525Gpa

Comparison of theoretical value vs experimental value

Percentage error = $((theoretical\ value - actual\ value)/theoretical\ value)\ x\ 100$

= (77Gpa - 3.0525Gpa)/ 77Gpa = 0.96 x 100 = 96% error

6. Descussion of results

The substantial 96% discrepancy observed between the theoretical and experimental values of the modulus of rigidity underscores significant differences that can occur in material characterization. This variation may be attributed to a variety of factors, including limitations inherent in experimental setups such as equipment precision, environmental conditions, and human error, as well as assumptions made in theoretical models regarding material properties and behavior under stress. Additionally, discrepancies may result from variations in sample composition and measurement techniques, underscoring the difficulties in accurately evaluating material properties. To address these disparities and enhance precision, it is crucial to enhance experimental techniques, validate theoretical assumptions, increase sample sizes, and conduct comprehensive error analyses. By tackling these issues, researchers can advance our comprehension of material behavior and lay the groundwork for more accurate material characterization methods

7. Conclusion

In summary, the torsion test is a key method for assessing how materials respond to torsional forces, which is essential for engineering purposes. Despite a notable 96% discrepancy between theoretical and experimental values of the modulus of rigidity, this research highlights the complexities and difficulties involved in material characterization. By acknowledging the limitations in experimental setups, theoretical models, and measurement techniques, and by employing strategies to improve accuracy such as refining experimental methods, validating theoretical assumptions, increasing sample sizes, and conducting thorough error analyses, we can enhance our comprehension of material behavior. These endeavors pave the way for more precise material characterization techniques, which are crucial for making well-informed decisions regarding material selection and engineering design.